Clique packings in graphs with forbidden subgraphs

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Abstract

For fixed $r \geq 3$, let $t_{r-1}(n)$ be the Turán number for K_r. We characterize the fixed graphs H for which every n-vertex H-free graph with $t_{r-1}(n)+k$ edges, where $k>0$, contains at least $(1-o(1))k$ edge-disjoint copies of K_r, uniformly over the excess k. This holds precisely when $\chi(H) \leq r+2$. The proof rests on a sharp finite fractional bound for K_{r+2}-free graphs. For triangles and each integer $M \geq 3$, we determine the best coefficient of $e(G)-n^2/4$ in a fractional packing lower bound for K_{M+1}-free graphs; it is $\min\{1, 2(M-1)/(3(M-2))\}$. This also gives the optimal asymptotic relative coefficient for every fixed forbidden graph of chromatic number at least four. We determine the sharp density curve for triangle packings in K_5-free graphs and, under the stronger K_{r+1}-free assumption, a nonlinear sharp density bound for K_r-packings. The latter has a finite integer form for K_4-free triangle packings. The proofs combine known symmetrization procedures with a reduction to the fractional matching polytope, an induction on the number of parts, and an optimization of the greedy-partition bound.

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1 Introduction

A basic packing problem asks how many edge-disjoint cliques a graph must contain in terms of its order and size. For a graph G, let $\nu_r(G)$ denote the maximum number of pairwise edge-disjoint copies of K_r in G. We ask which forbidden-subgraph conditions force the packing number to be asymptotically at least the excess above the Turán threshold, and how much this bound can be improved when a smaller clique is forbidden. All graphs are finite and simple. We write $T_j(n)$ for the balanced complete j-partite graph on n vertices and $t_j(n) = e(T_j(n))$. An H-free graph contains no subgraph isomorphic to H; the subgraph need not be induced. Turán's theorem places the threshold for forcing a K_r at $t_{r-1}(n)$.

Triangle packings have long been studied through edge decompositions. Erdős, Goodman, and Pósa [4] proved that every n-vertex graph can be decomposed into at most $t_2(n)$ triangles and single edges. Counting the members of such a decomposition gives $\nu_3(G) \geq (e(G)-t_2(n))/2$. Král', Lidický, Martins, and Pehova [12] later obtained a decomposition into edges and triangles whose total order is at most $n^2/2+o(n^2)$. Blumenthal, Lidický, Pehova, Pfender, Pikhurko, and Volec [3] determined the sharp bound on this minimum total order and characterized the extremal graphs for all sufficiently large n. In particular, their upper bound $n^2/2+1$ yields $\nu_3(G) \geq (2/3)(e(G)-t_2(n))-O(1)$ for arbitrary graphs.

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The K_4-free problem admits a stronger conclusion. Motivated by Erdős's question on clique decompositions with cost $|V(C)|-1$ for each member C, Győri [7, p. 360, Conjecture 4] recorded the conjecture that every K_4-free graph with $t_2(n)+k$ edges contains at least k edge-disjoint triangles. In this graph class every clique has order at most three, and a decomposition containing m triangles has cost $e(G)-m$, so the assertion is equivalent to a cost bound of $t_2(n)$. Huang and Shi [11] established quantitative partial bounds, including the conjectured bound in a range of excesses proportional to n^2. They also introduced a coloring argument that converts a triangle count into a packing by means of a partition into cliques. Győri and Keszegh [8] resolved the conjecture by combining this argument with a triangle-counting inequality for greedy clique partitions.

For general graphs, Balogh, He, Krueger, Nguyen, and Wigal [1, Theorems 1.4, 1.7, and 1.8(ii)] proved that every n-vertex graph admits, under this cost function, an edge decomposition of total cost at most $n^2/4+o(n^2)$, and a fractional edge decomposition of cost at most $t_2(n)$. The latter bound gives $\nu_3^*(G)\ge e(G)-t_2(n)$ when G is K_4-free. In a recent preprint, Zhang [15, Theorem 1.4] proved the exact integer cost bound $t_2(n)$ for all sufficiently large graphs.

For arbitrary graphs, Balogh and Wigal [2] proved that $e(G)=t_{r-1}(n)+k$ guarantees $\nu_r(G)\ge(2/r-o(1))k$ for every fixed $r\ge 3$. A theorem of Győri [6], in the form stated in [2, Theorem 1.6], gives the stronger estimate $\nu_r(G)\ge k-O_r(k^2/n^2)$ when $k=o(n^2)$. We determine exactly which fixed forbidden subgraphs guarantee the coefficient 1 uniformly over all positive excesses. For triangles, we also determine the optimal relative coefficient for every fixed forbidden graph of chromatic number at least four. Under the K_{r+1}-free assumption, we determine the sharper density bound, including a finite integer version for triangles.

A fractional K_r-packing assigns a nonnegative weight to each copy of K_r, with total weight at most 1 on every edge. We denote its maximum total weight by $\nu_r^*(G)$. Haxell and Rödl [9] proved that, for each fixed r, $\nu_r^*(G)-\nu_r(G)=o(n^2)$ uniformly over all n-vertex graphs; Yuster [14] extended this relation to packings by families of graphs. Thus fractional and integer packings have the same extremal density in the limit. A finite integer bound requires a separate argument, which we supply for triangles.

We normalize edge counts by n^2 and put $\rho_j=(j-1)/(2j)$, so that $t_j(n)=\rho_j n^2-O_j(1)$.

Theorem 1.1. *Fix an integer $r\ge 3$ and a graph H. The following property holds if and only if $\chi(H)\le r+2$. For every $\varepsilon>0$ there is an n_0 such that every H-free graph G on $n\ge n_0$ vertices with $k=e(G)-t_{r-1}(n)>0$ satisfies $\nu_r(G)\ge(1-\varepsilon)k$.*

The error in this statement is relative to the excess, even when the excess is much smaller than n^2. The fractional estimate underlying the classification is finite.

Theorem 1.2. *Every K_{r+2}-free graph G on n vertices, where $r\ge 3$, satisfies*

$$\nu_r^*(G)\ge e(G)-\rho_{r-1}n^2. \tag{1}$$

Equality holds for $T_{r+1}(n)$ whenever $r+1$ divides n. For $r=3$, the bound strengthens to $\nu_3^*(G)\ge e(G)-\lfloor n^2/4\rfloor$.

For triangles the linear bound determines the whole asymptotic density function in this graph class.

Corollary 1.3. *Let $0\le m_n\le t_4(n)$ be integers with $m_n/n^2\to p$. Then*

$$\lim_{n\to\infty}\frac{1}{n^2}\min_{\substack{v(G)=n,\ e(G)=m_n\\ G\text{ is }K_5\text{-free}}}\nu_3(G)=\max\{0,p-1/4\}.$$

The same conclusion holds for fractional packings.

For triangles, the linear bound extends to every forbidden clique with the best possible coefficient. For $M \geq 3$, put $\gamma_M = \min\{1, 2(M-1)/(3(M-2))\}$.

Theorem 1.4. *Let $M \geq 3$ be an integer. Every K_{M+1}-free graph G on n vertices satisfies*

$$\nu_3^*(G) \geq \gamma_M\big(e(G) - n^2/4\big).$$

For each $M \geq 3$, the coefficient γ_M is best possible.

A transfer to general forbidden subgraphs determines the optimal coefficient for any fixed forbidden graph of chromatic number at least four.

Corollary 1.5. *Let H be a fixed graph with $q = \chi(H) \geq 4$, and put $\beta_H = \gamma_{q-1}$. For every $\varepsilon > 0$ there is n_0 such that every H-free graph G on $n \geq n_0$ vertices with $e(G) = t_2(n) + k$, where $k > 0$, satisfies $\nu_3(G) \geq (\beta_H - \varepsilon)k$. The constant β_H cannot be increased.*

Forbidding K_{r+1} instead of K_{r+2} permits a stronger estimate at higher densities. We next give its exact asymptotic form throughout the feasible interval $0 \leq p \leq \rho_r$.

For $r \geq 3$, set $p_0 = \rho_{r-1} + 1/(4(r-1)^2)$ and, for $0 \leq p \leq \rho_r$, define

$$F_r(p) = \begin{cases} 0, & 0 \leq p \leq \rho_{r-1}, \\ p - \rho_{r-1}, & \rho_{r-1} \leq p \leq p_0, \\ \dfrac{\big(1 - \sqrt{r(r-2)(\rho_r - p)}\big)^2}{r^2}, & p_0 \leq p \leq \rho_r. \end{cases}$$

The two positive branches agree at p_0, where both have derivative 1.

Theorem 1.6. *Let $r \geq 3$ be an integer, and let G be a K_{r+1}-free graph on $n \geq 1$ vertices. With $p = e(G)/n^2$,*

$$\nu_r^*(G) \geq n^2 F_r(p). \tag{2}$$

The following finite bound also holds.

$$\nu_r^*(G) \geq e(G) - t_{r-1}(n). \tag{3}$$

For each fixed r, there is a function $\varepsilon_r(n) \to 0$ such that every such graph satisfies

$$\nu_r(G) \geq n^2 F_r(p) - \varepsilon_r(n) n^2. \tag{4}$$

Moreover, if m_n is any sequence of integers with $0 \leq m_n \leq t_r(n)$ and $m_n/n^2 \to p$, then

$$\lim_{n\to\infty} \frac{1}{n^2} \min_{\substack{|V(G)|=n,\ e(G)=m_n \\ G \text{ is } K_{r+1}\text{-free}}} \nu_r(G) = F_r(p). \tag{5}$$

The same limit holds with ν_r replaced by ν_r^*.

The fractional estimate is finite, whereas the integer estimate for general r has an asymptotic error. For triangles this error can be removed.

Theorem 1.7. *Every K_4-free graph G on $n \geq 1$ vertices satisfies*

$$\nu_3(G) \geq \left\lceil n^2 F_3\left(\frac{e(G)}{n^2}\right)\right\rceil. \tag{6}$$

In particular, if $5n^2/16 \leq e(G) \leq n^2/3$, then

$$\nu_3(G) \geq \left\lceil \frac{n^2}{9}\left(1-\sqrt{1-\frac{3e(G)}{n^2}}\right)^2\right\rceil.$$

For positive integer part sizes, equality holds in (6) *for every $K_{a,b,b}$ with $b \leq a \leq 2b$, and for every $K_{a,b,c}$ with $a = b + c$.*

Remark 1.8 (Comparison with known bounds). Győri and Keszegh [8] proved $\nu_3(G) \geq e(G) - \lfloor n^2/4 \rfloor$ for K_4-free graphs and asked for a strengthening above density 5/16. Theorem 1.7 gives its sharp density form, since

$$F_3(p) - \left(p - \frac{1}{4}\right) = \frac{\left(1-4\sqrt{1-3p}\right)^2}{36} > 0 \qquad (5/16 < p \leq 1/3).$$

This does not determine the exact extremal value for every finite pair (n, e); taking ceilings can remove a small real-valued improvement. Balogh and Wigal [2, Theorem 1.8] proved $\nu_r^*(G) \geq (2/r)(e(G) - \rho_{r-1}n^2)$ for arbitrary graphs. Theorem 1.6 determines the optimal density bound under the additional K_{r+1}-free assumption. Theorem 1.2 retains coefficient 1 when the forbidden clique is enlarged to K_{r+2}. It does not subsume the density refinement or the finite integer statement. For example, at $r = 3$ and $p = 1/3$, Theorem 1.6 gives packing density 1/9, whereas (1) gives 1/12.

The fractional proofs start from the symmetrization lemma of Balogh and Wigal [2], which allows an arbitrary coefficient of $e(G)$ in a linear objective involving $\nu_r^*(G)$. A general symmetrization theorem for weighted fractional clique covers and decompositions appears in [1, Theorem 1.7]. These replacements preserve the forbidden-clique condition and reduce the problem to complete multipartite graphs. The remaining task is to optimize the packing number under the resulting restriction on the number of parts.

For K_{r+2}-free graphs, the terminal graph can have $r+1$ parts. We normalize its fractional clique cover to obtain a point in the fractional matching polytope. The known half-integral description of its extreme points reduces the desired inequality to a quadratic estimate. For triangles, a zero coordinate of an extreme optimal cover gives a reduction that tracks the quadratic excess exactly. Induction on the number of parts then gives the optimal coefficient for every forbidden clique order. The theorem of Erdős, Frankl, and Rödl transfers the fractional bounds to general forbidden subgraphs. This initially gives an additive $o(n^2)$ error. Győri's small-excess estimate supplies the relative error when the excess is too small for that transfer to suffice.

Under the K_{r+1}-free assumption, the terminal graph has at most r parts. With exactly r nonempty parts, its fractional packing number is the product of its two smallest part sizes; with fewer parts, it contains no K_r. Optimizing the part sizes yields the supporting lines of F_r. The optimizing proportions also give constructions for every prescribed sequence of feasible edge counts.

For the finite triangle bound, we use the coloring lemma of Huang and Shi [11] together with the greedy-partition symmetrization of Győri and Keszegh [8]. Introducing a parameter into the triangle-counting objective gives a family of inequalities whose optimization recovers the same function F_3, without a rounding error. The resulting bound is attained by complete tripartite graphs. For larger cliques a finite integer analogue can fail, as illustrated by $K_{2,2,2,2}$ in Proposition 4.1.

Section 2 proves the fractional bounds and their density consequences. Section 3 proves the chromatic classification, the optimal relative coefficients for triangles, and the finite integer triangle bound. Section 4 discusses finite obstructions and further questions.

2 Fractional packings

We first prove the fractional bounds that underlie the chromatic classification and the optimal triangle coefficients. The common reduction gives a complete multipartite graph; its covering problem then yields the bound with one additional part and an induction for triangles with arbitrarily many parts. We finish by determining the sharper density curve when there are at most r parts.

Two nonadjacent vertices with the same neighborhood are called *clones*. We use the symmetrization lemma of Balogh and Wigal [2, Lemma 2.1] in the following form.

Corollary 2.1. *Let $r \geq 3$ and $s \geq 1$ be integers, let c be real, and let G be a K_{s+1}-free graph on n vertices. There is a complete multipartite graph H on the same vertex set, with at most s nonempty parts, such that*

$$\nu_r^*(H) + ce(H) \leq \nu_r^*(G) + ce(G).$$

Proof. For the objective $\nu_r^* + ce$, iterate the clone-class replacements of [2, Lemma 2.1], as in the proof of [2, Lemma 2.2], to obtain a complete multipartite graph without increasing the objective. Every replacement preserves K_{s+1}-freeness, since a clique uses at most one vertex of the copied independent class, and replacing that vertex by a source vertex gives a clique of the same size in the previous graph. Thus the resulting graph has at most s nonempty parts. □

We also use the following formula, given in [2, proof of Claim 3.2].

Lemma 2.2. *If $n_1 \geq \cdots \geq n_r > 0$, then*

$$\nu_r^*(K_{n_1,\ldots,n_r}) = n_{r-1}n_r.$$

A complete multipartite graph with fewer than r nonempty parts has fractional K_r-packing number zero.

The case of at most r parts has a finite linear bound. We record this consequence of the two known tools before treating $r+1$ parts.

Proposition 2.3. *Every K_{r+1}-free graph G on n vertices satisfies $\nu_r^*(G) \geq e(G) - t_{r-1}(n)$.*

Proof. Apply Corollary 2.1 with $c = -1$, and let H be the terminal graph. If H has at most $r-1$ parts, then $e(H) \leq t_{r-1}(n)$ and $\nu_r^*(H) = 0$. If H has r parts of sizes $n_1 \geq \cdots \geq n_r$, merge its two smallest parts. The resulting $(r-1)$-partite graph has $e(H) - n_{r-1}n_r = e(H) - \nu_r^*(H)$ edges, which is at most $t_{r-1}(n)$. In both cases $\nu_r^*(H) - e(H) \geq -t_{r-1}(n)$. The reduction transfers this inequality to G. □

We now allow $r+1$ parts. A fractional K_r-cover assigns nonnegative weights to the edges so that every K_r has total edge weight at least one. Linear programming duality identifies the minimum total cover weight with ν_r^*. In a complete multipartite graph, averaging over permutations within each part makes the cover constant on each pair of parts without changing its cost.

Proof of the general bound and equality in Theorem 1.2. Apply Corollary 2.1 with $s = r + 1$ and $c = -1$. For terminal graphs with at most r parts, Proposition 2.3 gives the required bound. Consider a complete $(r + 1)$-partite graph with positive part sizes $a_1, \dots, a_t$, where $t = r + 1$ and $\sum_i a_i = n$. Write w_{ij} for the cover weight on every edge between parts i and j, and put $W = \sum_{i<j} w_{ij}$. Each K_r omits exactly one part, so the cover constraints are $W - \sum_{j\neq i} w_{ij} \geq 1$ for every i. They force $W > 1$, since $W \leq 1$ would make all weighted degrees zero.

Set $z_{ij} = w_{ij}/(W - 1)$ and $\kappa = \sum_{i<j} z_{ij} = W/(W - 1) > 1$. Then z belongs to the fractional matching polytope

$$\mathcal{P} = \left\{ z \geq 0 : \sum_{j\neq i} z_{ij} \leq 1 \text{ for every } i \in [t] \right\},$$

and the cover cost is $\sum_{i<j} a_i a_j z_{ij}/(\kappa - 1)$. Let $L = e(G) - \rho_{r-1} n^2$. We may assume $L > 0$. It suffices to prove the affine inequality

$$\sum_{i<j} a_i a_j z_{ij} - L \left(\sum_{i<j} z_{ij} - 1 \right) \geq 0 \qquad (z \in \mathcal{P}). \tag{7}$$

The extreme points of $\mathcal{P}$ are half-integral. Their nonzero coordinates form vertex-disjoint edges of weight 1 and odd cycles of weight 1/2; see [13, Theorem 4]. It is enough to verify (7) at these points. If $\kappa \leq 1$, both terms on its left are nonnegative. Otherwise let I be the set of vertices incident to a nonzero coordinate, put $s = |I| = 2\kappa \geq 3$, and write $S = \sum_{i\in I} a_i$. Each vertex of I has weighted degree one. Since $L = n^2/(2(r - 1)) - \sum_i a_i^2/2$, inequality (7) is equivalent to

$$Q := \sum_i a_i^2 + \frac{4}{s-2} \sum_{i<j} z_{ij} a_i a_j \geq \frac{n^2}{t-2}.$$

When $s = 3$, the nonzero coordinates form a half-weight triangle, and the contribution to Q from I is S^2. For $s \geq 4$, that contribution equals

$$\frac{s-4}{s-2} \sum_{i\in I} a_i^2 + \frac{2}{s-2} \sum_{i<j} z_{ij} (a_i + a_j)^2.$$

The Cauchy–Schwarz inequality gives $\sum_{i\in I} a_i^2 \geq S^2/s$. It also gives $\sum_{i<j} z_{ij}(a_i + a_j)^2 \geq 2S^2/s$, because the sum of the weights is $s/2$ and $\sum_{i<j} z_{ij}(a_i + a_j) = S$. Thus in both cases the contribution from I is at least $S^2/(s - 2)$. If $s < t$, the remaining coordinates yield

$$Q \geq \frac{S^2}{s-2} + \frac{(n-S)^2}{t-s} \geq \frac{n^2}{t-2}.$$

If $s = t$, the same conclusion follows from $S = n$. This proves (7), so every fractional cover has cost at least L. Duality and the symmetrization reduction prove the bound.

When $r + 1$ divides n, every edge of $T_{r+1}(n)$ lies in the same positive number of copies of K_r. Uniform weights therefore give a fractional decomposition, and $\nu_r^*(T_{r+1}(n)) = e(T_{r+1}(n))/\binom{r}{2} = n^2/((r - 1)(r + 1)) = e(T_{r+1}(n)) - \rho_{r-1} n^2$. □

For triangles, an explicit four-partite calculation gives the finite rounding in Theorem 1.2 and the constructions for Corollary 1.3.

Lemma 2.4. *If $0 < a \leq b \leq c \leq d$, then $\nu_3^*(K_{a,b,c,d}) = \min\{ad + bc, ab + ac + bc\}$.*

Proof. Denote the parts by A, B, C, D in the indicated order. The edges between A, D together with those between B, C cover every triangle. So do the edges between the three parts A, B, C. These covers give the two upper bounds.

For the lower bound, prescribe the total weights of triangle types ABC, ABD, ACD, BCD, distributing each total uniformly within its type. If $d \geq b + c$, use the totals $(0, ab, ac, bc)$. The loads on AB, AC, BC equal their capacities, while those on AD, BD, CD are at most ad, bd, cd, respectively, since $a(b + c) \leq ad$, $b(a + c) \leq bd$ and $c(a + b) \leq cd$. The total weight is $ab + ac + bc$.

If $d \leq b + c$, put $x = a(b + c - d)/2$ and $y = a(b + d - c)/2$ and use $(x, y, ad - y, bc - x)$. These totals are nonnegative by the ordering of the part sizes. The loads on AB, AC, AD, BC equal their capacities. The other two loads are $a(d - c) + bc \leq bd$ and $a(d - b) + bc \leq cd$. This gives total weight $ad + bc$, as required. □

Proof of the triangle assertion in Theorem 1.2. For a complete four-partite graph as in Lemma 2.4,

$$e(G) - \nu_3^*(G) = \max\{(a + d)(b + c), d(n - d)\} \leq \lfloor n^2/4 \rfloor.$$

Graphs with at most three parts satisfy the same bound by Proposition 2.3. Corollary 2.1 with $s = 4$ and $c = -1$ transfers it to every K_5-free graph. □

Proof of Corollary 1.3. The lower bounds follow from Theorem 1.2, nonnegativity, and the Haxell–Rödl theorem. For $1/4 \leq p \leq 3/8$, take four part proportions $x, x, 1/2-x, 1/2-x$, where $0 \leq x \leq 1/4$. Their edge density is $p = 1/4 + x - 2x^2$, and Lemma 2.4 gives packing density $x - 2x^2 = p - 1/4$. These proportions cover the indicated interval continuously.

For a sequence with $1/4 \leq p < 3/8$, choose these proportions at density $\widehat{p}_n = \max\{m_n/n^2, 1/4\} + n^{-1/2}$ and round the part sizes. For all sufficiently large n the resulting K_5-free graph has $\widehat{p}_n n^2 + O(n) \geq m_n$ edges and fractional packing number $(\widehat{p}_n - 1/4)n^2 + O(n)$. Deleting edges to leave exactly m_n cannot increase either packing number. If $p < 1/4$, take any m_n edges of $T_2(n)$ for all sufficiently large n. At $p = 3/8$, delete $t_4(n) - m_n = o(n^2)$ edges from $T_4(n)$. These constructions prove the required upper limits in all cases. □

We next prove the optimal triangle coefficient for every forbidden clique order. If an extreme optimal cover has a zero coordinate, we remove the smaller of the two corresponding parts and the same number of vertices from the larger part. The covering constraints through that pair control exactly the change in the quadratic excess. If all coordinates are positive, the active constraints determine the cover and the edge count gives the bound directly.

Proof of Theorem 1.4. By Corollary 2.1, it suffices to consider complete multipartite graphs with at most M parts. We prove the bound for these graphs by induction on M. The case $e(G) \leq n^2/4$ follows from nonnegativity, so write $k = e(G) - n^2/4 > 0$.

For $M = 3$, Lemma 2.2 gives $\nu_3^*(K_{a,b,c}) = bc$ when $a \geq b \geq c$, and $e(K_{a,b,c}) - bc = a(b+c) \leq n^2/4$. Graphs with at most two parts have no positive excess. Suppose now that $M \geq 4$. Since $\gamma_{M-1} \geq \gamma_M$, the induction hypothesis handles graphs with fewer than M parts. Let $G = K_{a_1,\ldots,a_M}$, where all part sizes are positive, and write $e = e(G)$.

Averaging a fractional triangle cover within each part and applying linear programming duality, we obtain

$$\nu_3^*(G) = \min\left\{ \sum_{i<j} a_i a_j w_{ij} : w_{ij} \geq 0, \quad w_{ij} + w_{i\ell} + w_{j\ell} \geq 1 \text{ for all distinct } i, j, \ell \right\}.$$

All objective coefficients are positive, so an optimal cover w can be chosen at a vertex of this polyhedron. If all its coordinates are positive, the tight triangle constraints have full column rank. The constant vector with coordinates $1/3$ satisfies every triangle equality, and hence $w_{ij} = 1/3$ for all $i < j$. The covering cost is then $e/3$, and the bound $e \le (M-1)n^2/(2M)$ gives

$$\frac{e}{3} - \gamma_M \left(e - \frac{n^2}{4} \right) = \frac{(M-1)n^2/2 - Me}{3(M-2)} \ge 0.$$

Otherwise, relabel so that $w_{12} = 0$ and $a_1 = x \ge a_2 = y$, and put $S = \sum_{i=3}^{M} a_i$. For each $i \ge 3$, the triangle constraint through $1, 2, i$ gives $w_{1i} + w_{2i} \ge 1$. Form G' by deleting the second part and reducing the first part to size $x - y$, omitting it when $x = y$. The inherited weights form a fractional cover of G'. Writing W and W' for the two covering costs, we obtain

$$W = y \sum_{i=3}^{M} a_i (w_{1i} + w_{2i}) + W' \ge yS + \nu_3^*(G').$$

The graph G' has $n - 2y$ vertices and $e - xy - 2yS$ edges. Since $n = x + y + S$, its excess over the quadratic Turán term is $k - yS$. Apply the induction hypothesis with $B = \gamma_{M-1}$; this hypothesis also holds when that excess is nonpositive. As $B \le 1$ and $B \ge \gamma_M$, it follows that

$$W \ge yS + B(k - yS) = Bk + (1 - B)yS \ge \gamma_M k.$$

This proves the bound in both cases.

For $M \ge 4$, equality holds in balanced $T_M(n)$ whenever M divides n, since a uniform fractional triangle decomposition gives $\nu_3^*(T_M(n)) = e(T_M(n))/3$. For $M = 3$, the graph $K_{2t,t,t}$ has $\nu_3^* = t^2 = e - (4t)^2/4$. Thus each coefficient is best possible. □

The preceding results give sharp linear bounds. Under the stronger K_{r+1}-free assumption, we can also optimize at each fixed edge density. Corollary 2.1 and Lemma 2.2 reduce this task to the following family of supporting lines.

Proposition 2.5. *For $r \ge 3$ and $\lambda \ge 1$, put $D = r - 2 + r\lambda$ and $B_r(\lambda) = \lambda((r-3) + (r-1)\lambda)/(2D)$. Every K_{r+1}-free graph G on n vertices satisfies*

$$\nu_r^*(G) \ge \lambda e(G) - B_r(\lambda) n^2. \tag{8}$$

For $\lambda > 1$, equality in the continuous optimization is attained at the part proportions

$$\underbrace{\frac{1+\lambda}{D}, \ldots, \frac{1+\lambda}{D}}_{r-2 \text{ parts}}, \quad \frac{\lambda}{D}, \quad \frac{\lambda}{D}. \tag{9}$$

Proof. Apply Corollary 2.1 with $c = -\lambda$. It is enough to prove (8) for complete multipartite graphs with at most r nonempty parts. If there are at most $r - 1$ parts, then $e(G) \le \rho_{r-1} n^2$, and

$$B_r(\lambda) - \lambda \rho_{r-1} = \frac{\lambda(\lambda - 1)}{2(r-1)D} \ge 0.$$

This proves the required bound in that case.

Suppose there are r parts. Normalize their sizes by n, write y, z for the two smallest proportions, and put $s = y + z$. Denote the other $r - 2$ proportions by $x_1, \ldots, x_{r-2}$. Lemma 2.2 gives $\nu_r^*(G)/n^2 =$

yz. Since $\sum x_i = 1 - s$, the Cauchy–Schwarz inequality gives $\sum_{i<j} x_i x_j \le (r-3)(1-s)^2/(2(r-2))$. Also, $yz \le s^2/4$ and $1 - \lambda \le 0$. Hence

$$\begin{aligned}\frac{\nu_r^*(G) - \lambda e(G)}{n^2} &= (1-\lambda)yz - \lambda s(1-s) - \lambda \sum_{i<j} x_i x_j \\ &\ge \frac{1-\lambda}{4}s^2 - \lambda s(1-s) - \frac{\lambda(r-3)}{2(r-2)}(1-s)^2 = \frac{D}{4(r-2)}\left(s - \frac{2\lambda}{D}\right)^2 - B_r(\lambda).\end{aligned}$$

The nonnegative square proves (8). When $\lambda > 1$, equality in the two inequalities and in the square is attained by (9). These proportions are ordered as required, since $1 + \lambda > \lambda$. □

Optimizing the supporting lines gives the density function, and fractional rounding then yields the asymptotic integer bound.

Proof of the lower bounds in Theorem 1.6. The bound $\nu_r^*(G) \ge 0$ handles $p \le \rho_{r-1}$. At $\lambda = 1$, Proposition 2.5 gives $B_r(1) = \rho_{r-1}$ and thus $\nu_r^*(G) \ge n^2(p - \rho_{r-1})$. For $p_0 \le p < \rho_r$, put

$$z = \sqrt{r(r-2)(\rho_r - p)}, \qquad \lambda = \frac{(r-2)(1-z)}{rz}. \tag{10}$$

The condition $p \ge p_0$ is equivalent to $z \le (r-2)/(2(r-1))$, so $\lambda \ge 1$. Substituting (10) into (8) gives

$$\lambda p - B_r(\lambda) = \frac{(1-z)^2}{r^2} = F_r(p).$$

At $p = \rho_r$, taking $\lambda \to \infty$ in (8) gives $\nu_r^*(G) \ge n^2/r^2$. This proves (2) on the entire feasible interval. Proposition 2.3 proves (3).

The theorem of Haxell and Rödl [9], applied to K_r, states that for each fixed r and each $\varepsilon > 0$, every sufficiently large graph G on n vertices satisfies $\nu_r^*(G) - \nu_r(G) \le \varepsilon n^2$. The threshold is uniform over G. Applying this estimate to (2) proves (4). □

To prove sharpness, we describe complete multipartite graphs attaining every branch of the density function. Rounding their part sizes then gives the statement for every feasible sequence of edge counts, rather than only for rational densities along subsequences.

For $p \le \rho_{r-1}$, complete $(r-1)$-partite graphs with proportions $1 - (r-2)x, x, \dots, x$, where $0 \le x \le 1/(r-1)$, realize every density between 0 and ρ_{r-1} and contain no K_r. For $\rho_{r-1} \le p \le p_0$, take $r-2$ parts of proportion $1/(r-1)$ and two further parts of proportions y, z with $y + z = 1/(r-1)$ and $yz = p - \rho_{r-1}$. The conditions on p ensure that such nonnegative y, z exist. They are the two smallest proportions. The edge density is $\rho_{r-1} + yz = p$, and Lemma 2.2 gives fractional packing density $yz = F_r(p)$; a zero part corresponds to a K_r-free graph.

For $p_0 \le p < \rho_r$, use (9) with the parameter in (10). The last two proportions are $(1-z)/r$. The other $r-2$ proportions are $(1 + 2z/(r-2))/r$, so their squared sum gives edge density $\rho_r - z^2/(r(r-2)) = p$. The fractional packing density is $(1-z)^2/r^2 = F_r(p)$. At $p = \rho_r$, all r proportions are $1/r$.

Proof of the sharpness assertion in Theorem 1.6. Fix a feasible sequence m_n with $m_n/n^2 \to p$. The lower bounds already proved and the continuity of F_r give the required lower limit for both packing parameters. It remains to construct graphs giving the upper limit.

If $p < \rho_{r-1}$, then $m_n \le t_{r-1}(n)$ for all sufficiently large n. Choose any m_n edges of $T_{r-1}(n)$. The resulting graph has no K_r and therefore has both packing parameters equal to zero.

Suppose $\rho_{r-1} \le p < \rho_r$, and set $\widehat{p}_n = \max\{m_n/n^2, \rho_{r-1}\} + n^{-1/2}$. For all sufficiently large n, this lies in $[\rho_{r-1}, \rho_r)$ and tends to p. Choose the part proportions just described for density $\widehat{p}_n$, and round their multiples by n to nonnegative integers with sum n. Each part changes by at most a constant depending only on r. The resulting complete multipartite graph H_n has

$$e(H_n) = \widehat{p}_n n^2 + O_r(n), \qquad \nu_r^*(H_n) = F_r(\widehat{p}_n)n^2 + O_r(n).$$

The first quantity is at least m_n for all sufficiently large n, because the added density contributes $n^{3/2}$ edges. Delete $e(H_n) - m_n = o(n^2)$ edges. This does not introduce a K_{r+1} and cannot increase either packing parameter. The resulting graph G_n satisfies

$$\nu_r(G_n) \le \nu_r^*(G_n) \le F_r(\widehat{p}_n)n^2 + O_r(n) = \big(F_r(p) + o(1)\big)n^2.$$

This proves the upper limit in the interior, including $p = \rho_{r-1}$ and $p = p_0$.

Finally, if $p = \rho_r$, delete $t_r(n) - m_n = o(n^2)$ edges from $T_r(n)$. Lemma 2.2 gives $\nu_r^*(T_r(n)) = n^2/r^2 + O_r(n)$. Monotonicity under edge deletion therefore gives the upper limit $F_r(\rho_r) = 1/r^2$. This completes the proof of (5) and its fractional counterpart. □

3 Integer packings

The fractional estimates give integer bounds with an additive $o(n^2)$ error. To obtain the relative error in the chromatic classification and the optimal triangle coefficients, we combine them with the known small-excess estimate. We then prove the finite triangle bound under the stronger K_4-free assumption by a separate counting argument.

Proof of Theorem 1.1. Suppose first that $\chi(H) \le r + 2$, and choose a fixed complete $(r+2)$-partite graph F containing H. Every H-free graph is F-free. The theorem of Erdős, Frankl, and Rödl [5, Theorem 1.5] therefore gives a function $a(n) = o(n^2)$ such that deleting at most $a(n)$ edges from G produces a K_{r+2}-free graph G', uniformly over the graph class.

Let $b(n) = o(n^2)$ be the uniform error in the Haxell–Rödl theorem, and put $g(n) = \rho_{r-1}n^2 - t_{r-1}(n) = O_r(1)$. Theorem 1.2 gives

$$\nu_r(G) \ge \nu_r(G') \ge \nu_r^*(G') - b(n) \ge k - a(n) - b(n) - g(n).$$

Choose $h(n) = o(n^2)$ such that $a(n) + b(n) + g(n) = o(h(n))$. When $k \ge h(n)$, the preceding estimate gives $\nu_r(G) \ge (1 - o(1))k$ uniformly. When $0 < k < h(n)$, Győri's theorem [6], as stated in [2, Theorem 1.6], gives $\nu_r(G) \ge k - O_r(k^2/n^2) = (1 - o(1))k$ uniformly as well. The two ranges yield the asserted ε, n_0 statement.

Conversely, suppose $\chi(H) \ge r + 3$ and take $G = T_{r+2}(n)$ along integers divisible by $r + 2$. This graph is H-free and has a fractional K_r-decomposition by uniform weights. Hence

$$\frac{\nu_r(G)}{e(G) - t_{r-1}(n)} \le \frac{e(G)/\binom{r}{2}}{e(G) - t_{r-1}(n)} \longrightarrow \frac{2(r+1)}{3r} < 1.$$

Choosing $\varepsilon < 1 - 2(r+1)/(3r)$ disproves the property. □

Proof of Corollary 1.5. The theorem of Erdős, Frankl, and Rödl [5, Theorem 1.5] allows us to delete at most $a(n) = o(n^2)$ edges from G to obtain a K_q-free graph G', uniformly over all H-free graphs. Theorem 1.4 and the Haxell–Rödl theorem give

$$\nu_3(G) \ge \beta_H k - E(n), \qquad E(n) = o(n^2).$$

Choose $h(n) = o(n^2)$ with $E(n) = o(h(n))$. For $k \geq h(n)$ this gives $\nu_3(G) \geq (\beta_H - o(1))k$ uniformly. For $0 < k < h(n)$, Győri's theorem [6], in the form stated in [2, Theorem 1.6], gives $\nu_3(G) \geq k - O(k^2/n^2) = (1 - o(1))k$. Since $\beta_H \leq 1$, the two ranges establish the asserted uniform bound.

For sharpness when $q \geq 5$, use balanced $T_{q-1}(n)$ along integers divisible by $q-1$. These graphs are H-free, and

$$\frac{\nu_3(T_{q-1}(n))}{e(T_{q-1}(n)) - t_2(n)} \leq \frac{e(T_{q-1}(n))/3}{e(T_{q-1}(n)) - t_2(n)} \longrightarrow \gamma_{q-1}.$$

When $q = 4$, the H-free graph $K_{2t,t,t}$ has excess t^2 and fractional triangle packing number t^2, so the coefficient cannot exceed 1. □

For the finite triangle result, a coloring lemma converts a suitable lower bound on the total number of triangles directly into an integer packing. We obtain that lower bound by inserting a parameter into the greedy-partition symmetrization of Győri and Keszegh [8].

Definition 3.1. A *greedy clique partition* of a graph G with at least one vertex is a partition $P = (C_0, \ldots, C_h)$ of $V(G)$ into nonempty cliques such that $|C_0| \leq \cdots \leq |C_h|$ and $G[C_0 \cup \cdots \cup C_i]$ is $K_{|C_i|+1}$-free for every i. Its size is $q(P) = h + 1$.

Every graph with at least one vertex has such a partition. Repeatedly remove a maximum clique from the remaining graph and then order the removed cliques in reverse order. The maximum-clique condition gives both the size order and the forbidden-clique condition in the definition.

Write $t(G)$ for the total number of triangles in G. We use the coloring lemma of Huang and Shi [11], in the form stated in [8, arXiv version, Lemma 4].

Lemma 3.2 (Huang–Shi)**.** *If G is K_4-free and P is a greedy clique partition of G, then*

$$\nu_3(G) \geq \frac{t(G)}{q(P)}.$$

The next lemma supplies all the supporting lines needed for the finite triangle bound. At $\lambda = 1$ it is the triangle-count estimate of Győri and Keszegh [8, arXiv version, Theorem 5]. The proof follows their algorithm, with the parameter and its positive-region condition made explicit. Their concluding remarks also describe the use of related potential functions.

Lemma 3.3. *Let G be a K_4-free graph on $n \geq 1$ vertices, let P be a greedy clique partition, and let $\lambda \geq 1$. Then*

$$t(G) \geq q(P) \left(\lambda e(G) - \frac{\lambda^2}{1 + 3\lambda} n^2 \right).$$

Proof. Fix $\lambda \geq 1$, put $B = \lambda^2/(1+3\lambda)$, and suppose that $\Psi(G, P) = q(P)(\lambda e(G) - Bn^2) - t(G) > 0$. We use the generalized greedy partitions and the matching and merging procedure of [8, arXiv version, Section 2]. For such a partition, $q(P)$ is the size of the largest independent part of the first block plus the number of remaining clique blocks. The structural properties proved there ensure that each round absorbs the second block into the first, does not decrease $q(P)$, and preserves the partition requirements and the clique-number bound. We verify the change of objective needed here.

During one round, fix $q_0 = q(P)$. The objective $q_0(\lambda e - Bn^2) - t$ is linear in e and t. The fixed-parameter symmetrization in [8, arXiv version, Sections 2–3] therefore gives $q_0(\lambda e(G') - Bn^2) - t(G') > 0$ at the end of the round. Since $t(G') \geq 0$ and $q_0 > 0$, this implies $\lambda e(G') - Bn^2 > 0$. Thus replacing q_0 by $q(P') \geq q_0$ also does not decrease the objective, and the potential remains positive.

The procedure terminates at a complete multipartite graph. By K_4-freeness, write its part sizes as $a \geq b \geq c \geq 0$, with $a > 0$. Here $q = a$, $t = abc$, $e = a(n-a) + bc$, and $b + c = n - a$. Since $\lambda \geq 1$,

$$\begin{aligned}\frac{t}{q} - \lambda e &= (1-\lambda)bc - \lambda a(n-a) \geq \frac{1-\lambda}{4}(n-a)^2 - \lambda a(n-a) \\ &= \frac{1+3\lambda}{4}\left(a - \frac{1+\lambda}{1+3\lambda}n\right)^2 - Bn^2 \geq -Bn^2.\end{aligned}$$

The terminal potential is therefore nonpositive, a contradiction. □

Proof of the lower bound in Theorem 1.7. Choose a greedy clique partition P of G. Lemmas 3.2 and 3.3 give, for every $\lambda \geq 1$,

$$\nu_3(G) \geq \frac{t(G)}{q(P)} \geq \lambda e(G) - \frac{\lambda^2}{1+3\lambda}n^2.$$

The supporting-line coefficient here is exactly $B_3(\lambda)$. Taking $\lambda = 1$ gives $e(G) - n^2/4$, and choosing (10) with $r = 3$ gives the upper branch of $n^2 F_3(e(G)/n^2)$. The endpoint $p = 1/3$ follows by taking $\lambda \to \infty$. Together with $\nu_3(G) \geq 0$, these bounds cover all feasible densities. Since $\nu_3(G)$ is an integer, taking the ceiling proves (6). □

For the equality cases, we use the standard identity for positive integers $a \geq b \geq c$

$$\nu_3(K_{a,b,c}) = bc. \tag{11}$$

It follows from Lemma 3.2 and the fact that a greedy partition of $K_{a,b,c}$ has a cliques [8, arXiv version, Observation 7], together with the upper bound from the edges between the two smallest parts.

Proof of the equality assertions in Theorem 1.7. Consider $K_{a,b,b}$ with $b < a \leq 2b$. Set $\lambda = b/(a-b) \geq 1$. Its normalized part sizes agree with (9) for $r = 3$, so equality holds in the supporting line and in the optimized density bound. Identity (11) gives $\nu_3 = b^2 = n^2 F_3(e/n^2)$. If $a = b$, the graph is balanced and the same equality follows at $p = 1/3$.

If $a = b + c$, reorder b, c so that $b \geq c$. Then $a = n/2$, $e = n^2/4 + bc$, and $bc \leq n^2/16$. This density belongs to the linear branch, and (11) gives $\nu_3 = bc = n^2 F_3(e/n^2)$. In each case the expression before taking the ceiling is already an integer. □

4 Concluding remarks

Theorem 1.1 identifies the forbidden-subgraph conditions that force the full asymptotic excess bound. For triangles, Corollary 1.5 determines the best relative coefficient for every fixed forbidden graph of chromatic number at least four. The complete fixed-density function for K_{M+1}-free graphs remains open when $M \geq 5$. For triangles, Corollary 1.3 also determines the entire density function for K_5-free graphs. For $r \geq 4$, the corresponding density function for K_{r+2}-free graphs remains open; equality at $T_{r+1}(n)$ in Theorem 1.2 does not settle the intermediate densities. Even in the fractional setting, the continuous Turán term in that theorem cannot always be replaced by the finite one, since $G = K_5$ has $\nu_4^*(G) = 5/3 < 2 = e(G) - t_3(5)$. The same distinction is necessary in Theorem 1.4, since K_5 has $\nu_3^* = 10/3 < (8/9)(10 - t_2(5)) = 32/9$.

For integer triangle packings, K_4 already shows that $\nu_3(G) \geq e(G) - t_2(n)$ fails for K_5-free graphs. More generally, let $G = T_4(4a)$ with a odd. Every vertex has odd degree, whereas the edges

covered by triangles have even degree at each vertex. At least $n/2$ edges must remain uncovered, giving $\nu_3(G) \le e(G)/3 - n/6 = e(G) - n^2/4 - n/6$. Thus a uniform finite extension of the asymptotic bound would require an error of at least linear order. Whether an $O(n)$ error always suffices is a natural further question.

The finite triangle bound raises the question of whether the asymptotic error in Theorem 1.6 can also be removed for larger cliques. The following example shows that the analogous finite integer statement fails already for $r = 4$.

Proposition 4.1. *The graph $G = K_{2,2,2,2}$ is K_5-free and satisfies*

$$|V(G)| = 8, \qquad e(G) = 24, \qquad \nu_4(G) = 2, \qquad \nu_4^*(G) = 4.$$

In particular, neither $\nu_4(G) \ge n^2 F_4(e(G)/n^2)$ nor $\nu_4(G) \ge e(G) - t_3(n)$ holds for every K_5-free graph.

Proof. Label the two vertices in each part by 0 and 1. Each K_4 corresponds to a word in $\{0,1\}^4$. Two such cliques are edge-disjoint if and only if the words agree in at most one coordinate, or equivalently have Hamming distance at least 3.

Suppose three words have pairwise distance at least 3. Flipping coordinates makes the first word 0000 without changing the distances. Each of the other two words then has at least three entries equal to 1, so their distance is at most 2, a contradiction. The words 0000 and 1111 give a packing of size 2.

Lemma 2.2 gives $\nu_4^*(G) = 4$. Finally, $e(G)/n^2 = 3/8 = \rho_4$, hence $n^2 F_4(e(G)/n^2) = 4$, whereas $e(G) - t_3(8) = 24 - 21 = 3$. Both proposed integer bounds exceed $\nu_4(G) = 2$. □

Theorem 1.6 determines the leading term at each fixed feasible density, and Theorem 1.7 gives a finite bound attained by explicit families. Further questions include a characterization of equality in (6), a stability theorem above the transition density, and the smallest possible integer error for $r \ge 4$.

A separate question concerns a specified greedy partition P of size q. He, Ma, Wang, and Zu [10] disproved the proposed triangle-count inequality $t(G) \ge q(e(G) - q(n-q))$ from [8]. As they note in their concluding remarks, the corresponding packing assertion $\nu_3(G) \ge e(G) - q(n-q)$ remains open. Our bound depends only on n and $e(G)$ and does not establish that partition-dependent assertion.

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